BEYOND BROWN'S RADIAL TESTS: ADMISSIBILITY, PARABOLICITY, AND A DOUBLING–POINCARÉ PRIOR THAT NEITHER TEST DECIDES

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Abstract. For the estimation of a normal mean in $\mathbb{R}^m$ under quadratic loss, Brown (1971) proved that a generalized Bayes estimator δ_F is admissible if and only if a diffusion attached to the marginal f^* of the prior F is recurrent, and gave two radial tests: a sphere-average test for recurrence and a ray test for transience. He observed that both tests can be silent. We read his theory as the potential theory of the weighted Euclidean space $(\mathbb{R}^m, f^*\,\mathrm{d}x)$ and ask what the modern theory of recurrence on weighted manifolds adds inside the zone where neither test applies. Three results. (A) If $f^*\,\mathrm{d}x$ is doubling and supports a scale-invariant Poincaré inequality, the sphere test is already complete: recurrence, Brown's sphere test and the ball-volume test are equivalent, so geometric criteria can add decidable cases only on the inadmissible side. (B) They do add one. An explicit prior on $\mathbb{R}^3$ whose marginal is small along a lattice of circles at dyadic scales with spacing equal to the square root of the scale, is doubling and Poincaré, silences both of Brown's tests, and is transient by the ball-volume test; its generalized Bayes estimator is inadmissible. (C) Outside the doubling–Poincaré class the zone is two-sided: the same circles placed on one surface per scale give an admissible estimator, decided by a flux bound along a non-spherical exhaustion, and a logarithmic product prior gives an inadmissible one, decided by product heat kernels. A comparison lemma shows that any positive weight with Lipschitz logarithm is, up to constants, the marginal of a prior, so that explicit weights may be used throughout.

1. Introduction

Let $X \sim N_m(\theta, I_m)$ and consider estimators of θ under quadratic loss. For a σ-finite measure F on $\mathbb{R}^m$ with finite Gaussian marginal $f^* = \varphi * F$, where φ is the standard normal density, the generalized Bayes estimator is

$$\delta_F(x) = x + \nabla \log f^*(x).$$

Brown [4] showed that every admissible estimator is of this form and characterized admissibility by the behaviour of a diffusion: let $\{Z_t\}$ be the diffusion on $\mathbb{R}^m$ with generator $\Delta + \nabla \log f^* \cdot \nabla$, the weighted Laplacian of the measure $\mu = f^*\,\mathrm{d}x$. Then δ_F is inadmissible if $\{Z_t\}$ is transient, and admissible if $\{Z_t\}$ is recurrent and the drift $\nabla \log f^*$ is bounded. Since the recurrence of $\{Z_t\}$ is the *parabolicity* of the weighted Euclidean space $W_F = (\mathbb{R}^m, \delta_{ij}, f^*\,\mathrm{d}x)$, Brown's theorem places the admissibility problem inside the potential theory of weighted manifolds, a subject that has developed independently since [20, 10, 8]. The statistical literature has long used the Dirichlet-form formulation of Brown's criterion [14, 13, 6, 25]; the geometric reading, and the tests it brings, are the subject of this note.

Brown gave two tests that depend only on radial information. Write $\bar{f}(r)$ for the mean of f^* over the sphere $\{|x| = r\}$. His sphere test says that $\int_1^\infty \mathrm{d}r/(r^{m-1}\bar{f}(r)) = \infty$ implies recurrence; his ray test says that if the ray integrals $\int_1^\infty \mathrm{d}r/(r^{m-1}f^*(r\varphi))$ are uniformly bounded on a set of directions φ of positive measure, then δ_F is inadmissible. He remarked [4, p. 900] that there are marginals on which neither test applies. We call this set of marginals *Brown's undecided zone*.

The geometric literature knows the sphere test as the sphere-area parabolicity test [8, Thm. 7.5], and knows further tests with no statistical counterpart: the ball-volume test of Karp, Varopoulos

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and Grigor'yan, which is an equivalence when the measure is doubling and supports a Poincaré inequality [8, Thm. 11.1], and the flux test along an arbitrary exhaustion [8, Thm. 7.6]. Our question is what these add inside Brown's undecided zone. The answer has three parts.

(A) *Inside the doubling–Poincaré class, Brown's sphere test is complete* (Theorem 3.1). If $f^*\,\mathrm{d}x$ is doubling and supports a scale-invariant Poincaré inequality, then recurrence, the divergence of the sphere integral, and the divergence of the volume integral $\int^\infty r\,\mathrm{d}r/\mu(B_r)$ are equivalent. So for such priors with bounded drift, δ_F is admissible if and only if Brown's 1971 test says so, and geometric criteria can only decide cases on the inadmissible side.

(B) *They do decide one* (Theorem 4.1). We construct a prior on $\mathbb{R}^3$ whose marginal is doubling and Poincaré, on which both of Brown's tests are silent (every ray integral diverges, the sphere integral converges), and whose volume grows like r^3; the ball-volume test shows transience and δ_F is inadmissible. The construction places circles of radius $\asymp 2^k$ in the shell $2^k \le |x| < 2^{k+1}$ at mutual distance $\asymp 2^{k/2}$, so that their angular positions cover every direction while no coherent surface of small weight forms.

(C) *The zone is two-sided* (Section 5). Placing the same circles on one surface per shell produces a prior on which both tests are silent and which is nevertheless recurrent, hence admissible; the proof is a flux bound along the surfaces. A logarithmic product prior is transient, hence inadmissible, by product heat kernels. Neither prior is doubling and Poincaré, and Theorem 3.1 says why the first cannot be.

A comparison lemma (Lemma 2.3) shows that any positive weight w with Lipschitz logarithm satisfies $c\,w \le \varphi * w \le C\,w$ with bounded drift, so that every explicit weight below is, up to constants, the marginal of the prior $F = w\,\mathrm{d}\theta$. All statements about recurrence, doubling, Poincaré inequalities and Brown's tests are stable under such comparability.

The examples of Section 5 connect to the question of Johnstone and Lalley [14], whether any generalized Bayes estimator that is not proper Bayes is immune to the Stein effect under products; their theory decides the product example and our construction in Theorem 4.1 is the non-product counterpart. The same dichotomy between parabolic and non-parabolic geometry governs Komaki's theory of shrinkage priors for predictive densities, where a prior asymptotically dominating the Jeffreys prior exists if and only if the Fisher manifold of the model is non-parabolic [17]; Appendix A records that parallel and the identity that expresses Komaki's risk reduction through Perelman's weighted scalar curvature.

Organization. Section 2 fixes notation, restates Brown's theorems in geometric language, proves the comparison lemma, and records the volume criterion. Section 3 proves Theorem 3.1 and locates the undecided zone through the spherical A_2 ratio. Section 4 constructs the prior of Theorem 4.1 and proves its properties. Section 5 gives the two further examples. Section 6 reports numerical companions to the proofs, and Section 7 discusses what the geometric criteria add and what remains open.

2. Preliminaries

2.1. **Notation.** Throughout, $m \ge 2$, $B(x,\rho)$ is the open Euclidean ball, $B_r = B(0,r)$, and ω_{m-1} the area of the unit sphere. For a positive locally integrable weight w on $\mathbb{R}^m$ we write $\mu = w\,\mathrm{d}x$ and $V(r) = \mu(B_r)$. For $r \ge 1$ let $\bar f(r)$ and $A(r)$ be the means of w and of $1/w$ over the sphere $\{|x| = r\}$, and $Q(r) = \bar f(r)A(r) \ge 1$ (Jensen) the *spherical A_2 ratio*. The measure μ is *doubling* if $\mu(B(x,2\rho)) \le D\,\mu(B(x,\rho))$ for all balls, and supports the *scale-invariant Poincaré inequality* if

$$\int_B |u - u_B|^2\,\mathrm{d}\mu \;\le\; C_P\,\rho^2 \int_B |\nabla u|^2\,\mathrm{d}\mu \qquad \text{for every ball } B \text{ of radius } \rho \text{ and every } u \in C^1(\bar B), \tag{1}$$

where u_B is the μ-mean of u on B. We say that μ (or w) is $D{+}P$ if both hold. The best constant in (1) on a given ball is $1/(\lambda_1(B)\rho^2)$, where $\lambda_1(B)$ is the first nonzero Neumann eigenvalue of

the weighted Laplacian $-w^{-1}\operatorname{div}(w\nabla\cdot)$ on B; this is how the inequality is tested numerically in Section 6.

The weighted space $(\mathbb{R}^m, \delta_{ij}, \mu)$ carries the Dirichlet form $\mathcal{E}(u) = \int |\nabla u|^2 \,\mathrm{d}\mu$ and the weighted Brownian motion with generator $\Delta + \nabla \log w \cdot \nabla$. It is *parabolic* if the capacity of some (equivalently every) ball vanishes, $\operatorname{cap}(B_1) = \inf\{\mathcal{E}(j) : j \in C_c^\infty,\ j \geq 1 \text{ on } B_1\} = 0$, equivalently if the weighted Brownian motion is recurrent [8, Thm. 5.1]; otherwise it is *non-parabolic* or *transient*.

2.2. **Brown's theorems.** We restate the results of [4] that we use, in the notation above with $w = f^*$.

Theorem 2.1 (Brown [4])**.** *Let F be a σ-finite measure on $\mathbb{R}^m$ with $f^* = \varphi * F < \infty$, and let $\{Z_t\}$ be the weighted Brownian motion of $(\mathbb{R}^m, f^*\,\mathrm{d}x)$.*

(a) (Thm. 4.3.1) *$\{Z_t\}$ is transient if and only if $\operatorname{cap}(B_1) > 0$ in $(\mathbb{R}^m, f^*\,\mathrm{d}x)$.*

(b) (Thm. 5.1.1) *If $\{Z_t\}$ is transient, δ_F is inadmissible. If $\{Z_t\}$ is recurrent and $\sup|\nabla \log f^*| < \infty$, δ_F is admissible.*

(c) (Thm. 6.3.1) *If F is spherically symmetric and $\nabla \log f^*$ is bounded, δ_F is inadmissible if and only if*

$$\int_1^\infty \frac{\mathrm{d}r}{r^{m-1} f^*(r)} < \infty;$$

the boundedness is used only for the admissible direction, as in (b).

(d) (Thm. 6.4.3, sphere test) *If*

$$\int_1^\infty \frac{\mathrm{d}r}{r^{m-1} \bar{f}(r)} = \infty,$$

then $\{Z_t\}$ is recurrent; hence δ_F is admissible if moreover $\nabla \log f^*$ is bounded.

(e) (Thm. 6.4.4, ray test) *If a set Ω of directions of positive surface measure satisfies*

$$\sup_{\varphi \in \Omega} \int_1^\infty \frac{\mathrm{d}r}{r^{m-1} f^*(r\varphi)} < \infty,$$

then δ_F is inadmissible.

Brown's proofs are those of the corresponding geometric statements: (a) is the Dirichlet principle for the capacity of B_1; (d) uses the radial test functions $j_i(x) = c_i \int_{|x|}^i \mathrm{d}r/(r^{m-1}\bar{f}(r))$, whose energy tends to zero, exactly as in the sphere-area test of [8, Thm. 7.5]; (e) integrates a one-dimensional capacity bound along rays. We say that the sphere test is *silent* on F when its integral converges, and that the ray test is silent when no set of directions of positive measure has uniformly bounded ray integrals. Brown's undecided zone is the set of F on which both are silent. Table 1 collects the dictionary.

Remark 2.2 (priority). Grigor'yan [8, p. 178] attributes the sphere-area test for Riemannian manifolds to Lyons–Sullivan [20] and to [10]; its classical ancestor for Riemann surfaces is the Ahlfors–Nevanlinna criterion of the 1930s. Brown's Theorem 6.4.3 is the same test on the weighted space W_F, with the same proof, carried out in 1971 in the statistical setting and so predating the general weighted-manifold statements; his Theorem 4.3.1 is the capacity characterization of parabolicity. In the sources we have read, neither literature cites the other.

2.3. **Explicit weights are marginals.** Brown's theorems concern Gaussian marginals $f^* = \varphi * F$. The weights we construct are explicit functions of the distance to a set, which are not Gaussian marginals as they stand. The following lemma removes the distinction.

Lemma 2.3 (comparison)**.** *Let $w > 0$ on $\mathbb{R}^m$ be locally integrable with $\log w$ Lipschitz with constant L, and let $f^* = \varphi * w$, the marginal of the prior $F = w\,\mathrm{d}\theta$. Then*

(i) *$c_L\, w \leq f^* \leq C_L\, w$ with $c_L = E\, e^{-L|Z|} > 0$ and $C_L = E\, e^{L|Z|} < \infty$, $Z \sim N(0, I_m)$;*

TABLE 1. Brown's admissibility theory read as potential theory of $W_F = (\mathbb{R}^m, \delta, f^*\,\mathrm{d}x)$. Theorem numbers refer to [4] and [8].

statistics	weighted potential theory
generalized Bayes estimator $\delta_F = x + \nabla \log f^*$	weighted space W_F with measure $f^*\,\mathrm{d}x$
Brown's diffusion $\{Z_t\}$	weighted Brownian motion, generator $\Delta + \nabla \log f^* \cdot \nabla$
Brown's energy infimum (Thm. 4.3.1)	capacity of the unit ball (Grigor'yan, Thm. 5.1)
δ_F admissible, bounded drift (Thm. 5.1.1)	W_F parabolic
spherical F (Thm. 6.3.1)	model-manifold test $\int^\infty \mathrm{d}r/S(r)$, $S(r) = \omega_{m-1} r^{m-1} f^*(r)$
sphere test (Thm. 6.4.3)	sphere-area parabolicity test (Grigor'yan, Thm. 7.5)
—	ball-volume test $\int^\infty r\,\mathrm{d}r/V(r) = \infty$ (Thm. 7.3); an equivalence under D+P (Thm. 11.1)
—	flux test along an arbitrary exhaustion (Thm. 7.6); Lemma 5.1 below
MLE admissible iff $m \le 2$ (Stein [23])	Brownian motion recurrent iff $m \le 2$

(ii) $|\nabla \log f^*| \le L$;

(iii) *F is σ-finite and $f^* < \infty$, so Theorem 2.1 applies to δ_F;*

(iv) *the spaces $(\mathbb{R}^m, w\,\mathrm{d}x)$ and $(\mathbb{R}^m, f^*\,\mathrm{d}x)$ have comparable Dirichlet forms, hence the same parabolicity type; doubling constants, Poincaré constants, sphere means, ray integrals and ball volumes of the two weights are comparable, so Brown's tests are silent for w if and only if they are silent for f^*, and w is D+P if and only if f^* is.*

Proof. From $e^{-L|z|} \le w(x+z)/w(x) \le e^{L|z|}$, integration against φ gives (i); $\nabla f^* = \varphi * (w \nabla \log w)$ and $|\nabla \log w| \le L$ give (ii); (iii) is immediate, and (iv) follows from (i) applied to $\int |\nabla u|^2 w\,\mathrm{d}x$ and to the integrals that define the other quantities, together with Theorem 2.1(a). □

For $m = 3$ and $L = 2$ one has $C_L = 73.8$ and $c_L = 0.085$. Some of the weights below are marginals exactly: for $y, Z \in \mathbb{R}^2$, $E|y+Z|^4 = |y|^4 + 8|y|^2 + 8$, so the weight $(1+|y|^2)^2$ is, up to constants, the marginal of $\mathrm{dist}(\theta, \ell)^4\,\mathrm{d}\theta$ for a line $\ell \subset \mathbb{R}^3$.

2.4. **The volume criterion under doubling and Poincaré.** The following is the theorem of the geometric literature on which Theorem 3.1 rests.

Theorem 2.4 (Grigor'yan; Saloff-Coste; Sturm)**.** *Let w be D+P on $\mathbb{R}^m$. Then $(\mathbb{R}^m, w\,\mathrm{d}x)$ is parabolic if and only if*

$$\int_1^\infty \frac{r\,\mathrm{d}r}{V(r)} = \infty. \tag{2}$$

Sources. By Lemma 2.3 we may take w smooth, so that $-w^{-1}\,\mathrm{div}(w\nabla\cdot)$ is a uniformly elliptic operator with smooth coefficients, symmetric with respect to μ. For such operators Saloff-Coste proved that doubling together with the Poincaré inequality is equivalent to the parabolic Harnack inequality [22, Thm. 3.1] and implies two-sided Gaussian bounds for the heat kernel [22, Thm. 4.2]; see also [11] and, for local Dirichlet spaces, [24]. In particular the upper bound $p(t,x,x) \le C/V(x, \sqrt{t})$ holds, which together with doubling is equivalent to the relative Faber–Krahn inequality [8, remark after Thm. 11.1]. Under the relative Faber–Krahn inequality, non-parabolicity is equivalent to the convergence of the integral in (2) [8, Thm. 11.1]; the weighted-manifold form of these arguments is in [9]. The implication "(2) ⇒ parabolic" holds without any hypothesis [8, Thm. 7.3]. □

Corollary 2.5 (the sphere test dominates the volume test)**.** *For any positive weight, divergence of $\int^\infty r\,\mathrm{d}r/V(r)$ implies divergence of $\int^\infty \mathrm{d}r/(r^{m-1}\bar{f}(r))$.*

Proof. $V'(r) = \omega_{m-1} r^{m-1} \bar{f}(r)$. By the Cauchy–Schwarz inequality on $[R, 2R]$, $R^2 = \left(\int_R^{2R} \mathrm{d}r\right)^2 \le \int_R^{2R} V' \,\mathrm{d}r \cdot \int_R^{2R} \mathrm{d}r/V' \le V(2R) \int_R^{2R} \mathrm{d}r/V'$, while $\int_R^{2R} r \,\mathrm{d}r/V \le 2R^2/V(R)$. Summing over $R = 2^j$, divergence of the volume integral forces $\sum_j 4^j/V(2^{j+1}) = \infty$ and hence divergence of $\int \mathrm{d}r/V'$. □

3. The reach of the radial tests

Theorem 3.1 (the sphere test is complete under D+P)**.** *Let w be D+P on $\mathbb{R}^m$, $m \ge 2$. Then the following are equivalent:*

(i) $(\mathbb{R}^m, w\,\mathrm{d}x)$ *is parabolic;*
(ii) $\int_1^\infty \mathrm{d}r/(r^{m-1}\bar{f}(r)) = \infty$ *(Brown's sphere test fires);*
(iii) $\int_1^\infty r\,\mathrm{d}r/V(r) = \infty$ *(the ball-volume test fires).*

Consequently, if f^* is D+P with bounded drift, δ_F is admissible if and only if Brown's sphere test fires. The ray test implies transience but is not necessary for it: the weight of Theorem 4.1 is D+P and transient with the ray test silent.

Proof. (iii)⇔(i) is Theorem 2.4. (ii)⇒(i) is Brown's sphere test, Theorem 2.1(d). (i)⇒(iii) is Theorem 2.4 again, and (iii)⇒(ii) is Corollary 2.5. The statement about δ_F is Theorem 2.1(b) with Lemma 2.3 if f^* is only comparable to a D+P weight. The last sentence is Theorem 4.1. □

Thus, within the D+P class, the geometric literature adds nothing on the admissible side, where Brown's 1971 sphere test is necessary and sufficient, and it can add exactly one kind of case on the inadmissible side: a transient weight along whose every ray the escape to infinity is expensive. Theorem 4.1 shows that this case occurs. The next two lemmas locate the undecided zone more precisely and explain why one spatial dimension is not enough to enter it.

Lemma 3.2 (bounded spherical A_2 ratio)**.** *Suppose $\sup_{r\ge1} Q(r) = C < \infty$. If $\int_1^\infty \mathrm{d}r/(r^{m-1}\bar{f}(r)) = \infty$ then $(\mathbb{R}^m, w\,\mathrm{d}x)$ is parabolic; if the integral converges then the ray test fires. Hence Brown's two tests are exhaustive on weights with bounded spherical A_2 ratio, and every weight in the undecided zone has $Q(r) \to \infty$ along a sequence.*

Proof. The first case is Theorem 2.1(d). In the second, let $R(\varphi) = \int_1^\infty \mathrm{d}r/(r^{m-1}w(r\varphi))$. By Fubini its mean over the unit sphere is $\int_1^\infty A(r)\,\mathrm{d}r/r^{m-1} \le C\int_1^\infty \mathrm{d}r/(r^{m-1}\bar{f}(r)) < \infty$, so by Chebyshev's inequality the set $\{\varphi : R(\varphi) \le 2\,\mathrm{mean}\,R\}$ has at least half the surface measure, and R is bounded on it: the hypothesis of Theorem 2.1(e). □

Lemma 3.3 (one-dimensional D+P forces A_2)**.** *Let $v > 0$ be continuous on $\mathbb{R}$ with $\mu_1 = v\,\mathrm{d}t$ doubling, $\mu_1(2I) \le D\mu_1(I)$ for all intervals I, and satisfying $\int_I |u - u_I|^2\,\mathrm{d}\mu_1 \le C|I|^2 \int_I |u'|^2\,\mathrm{d}\mu_1$ for all intervals. Then, with $\nu_1 = v^{-1}\,\mathrm{d}t$, $\mu_1([-r,r])\,\nu_1([-r,r]) \le 32\,CD^3r^2$ for every $r > 0$. Consequently the product weight $w(x) = v(x_1)$ on $\mathbb{R}^m$, $m \ge 3$, has bounded spherical A_2 ratio, and δ_F is inadmissible for every such prior.*

Proof. Take $I = [-2r, 2r]$ and the test function $u = 0$ on $[-2r, -r]$, $u = 1$ on $[r, 2r]$, and $u(t) = \nu_1([-r,t])/\nu_1([-r,r])$ in between, which minimizes the energy across the middle segment: $\int_I |u'|^2\,\mathrm{d}\mu_1 = 1/\nu_1([-r,r])$. A function equal to 0 on a set of measure a and to 1 on a set of measure b has weighted variance at least $ab/(a+b) \ge \frac{1}{2}\min(a,b)$. Since $[-r,r]$ is contained in the eightfold dilates of $[r,2r]$ and of $[-2r,-r]$, doubling gives $\mu_1([r,2r]) \ge \mu_1([-r,r])/D^3$ and likewise on the left, and the Poincaré inequality reads $\mu_1([-r,r])/(2D^3) \le C(4r)^2/\nu_1([-r,r])$. For the product weight, the sphere means are weighted means of v and $1/v$ over $[-r,r]$ with a kernel bounded above and bounded below on $[-r/\sqrt{2}, r/\sqrt{2}]$, so by doubling $\bar{f}(r) \asymp \mu_1([-r,r])/r$ and $A(r) \lesssim \nu_1([-r,r])/r$, whence Q is bounded. Doubling also gives reverse doubling, $\mu_1([-r,r]) \ge cr^\varepsilon$, so $\int^\infty \mathrm{d}r/(r^{m-1}\bar{f}) \lesssim \int^\infty r^{2-m-\varepsilon}\,\mathrm{d}r < \infty$ for $m \ge 3$, and Lemma 3.2 gives inadmissibility. □

Both hypotheses are needed: $v(t) = (1+|t|)^2$ is doubling but fails the Poincaré inequality, because the valley at 0 is a bottleneck between two heavy sides, and $v(t) = (1+|t|)^{-2}$ satisfies the Poincaré inequality but is not doubling; in both cases $\mu_1\nu_1/r^2$ grows like r (Figure 2b,c). The mechanism of Lemma 3.3 is that a valley in one dimension has co-dimension one and disconnects the interval. In $\mathbb{R}^m$, $m \geq 3$, a valley along a curve has co-dimension two, does not disconnect, and the lemma fails:

Theorem 3.4 (co-dimension-two valleys)**.** *For $m \geq 3$ and $\alpha \geq 0$ the weight $w(x) = (1 + x_1^2 + \cdots + x_{m-1}^2)^{\alpha/2}$ on $\mathbb{R}^m$, a power of the distance to the x_m-axis, is D+P. For $\alpha > m-1$ its spherical A_2 ratio satisfies $Q(r) \asymp r^{\alpha-(m-1)} \to \infty$.*

Proof. Björn [3, Prop. 6], generalizing Chanillo and Wheeden, proves that for $x = (x', x'')$ with $x' \in \mathbb{R}^k$, $2 \leq k \leq n$, any $v \in A_p$, points $a_j \in \mathbb{R}^k$ and exponents $\gamma_0, \gamma_j \geq 0$, the weight $v(x)(1 + |x'|)^{\gamma_0} \prod_j \big(|x' - a_j|/(1 + |x' - a_j|)\big)^{\gamma_j}$ on $\mathbb{R}^n$ is p-admissible, that is, doubling and supporting the weak $(1,p)$-Poincaré inequality. Take $n = m$, $k = m-1$, $v \equiv 1$, $\gamma_0 = \alpha$, no a_j, and $p = 2$; since $(1 + |x'|)^\alpha \asymp (1 + |x'|^2)^{\alpha/2}$, the weight is doubling and supports the weak $(1,2)$-Poincaré inequality in the sense of [12, (5), p. 7], which under doubling self-improves to the strong $(2,2)$ form (1): [12, Thm. 5.1] raises the exponent on the left to any $h < sp/(s-p)$, with the energy on a dilated ball, and [12, Thm. 9.7] removes the dilation on any weak John domain, in particular on a ball (the exponent s in the lower mass bound exists for every doubling measure, [12, Lemma 14.6]). For the asymptotics parameterize the sphere $\{|x| = r\}$ by $|x'| = r\sin\theta$; then $\bar{f}(r) \asymp r^\alpha$, and for $\alpha > m-1$ the integral of $1/w$ concentrates on the polar caps $\theta \lesssim 1/r$, giving $A(r) \asymp r^{-(m-1)}$. □

So Lemma 3.3 is a one-dimensional statement, and weights in the undecided zone that are D+P must have valleys of co-dimension at least two. The ray test, however, is silent only if *every* direction (up to a null set) meets cheap regions at infinitely many scales, which a single line does not achieve. The construction of the next section makes the valleys sweep every direction without ever forming a surface.

4. A doubling–Poincaré prior that neither test decides

4.1. **The construction.** Work in $\mathbb{R}^3$ with spherical coordinates (r, θ, ϕ), $\theta \in [0, \pi]$ the polar angle. For $k \geq 4$ put

$$N_k = 2^{k+2}, \qquad L_k = 2^{\lceil k/2 \rceil}, \qquad M_k = N_k/L_k, \qquad s_k = 2^{k/2}.$$

Shell k is $\{2^k \leq |x| < 2^{k+1}\}$. For $0 \leq i < L_k$ and $0 \leq j < M_k$ let $n = jL_k + i$ and let $\Gamma_{k,i,j}$ be the latitude circle

$$\Gamma_{k,i,j} = \Big\{ |x| = 2^k + i\,\frac{2^k}{L_k},\ \ \theta = \theta_n := \frac{\pi(n+\frac{1}{2})}{N_k} \Big\}.$$

Since $(i,j) \mapsto n$ is a bijection onto $\{0, \dots, N_k - 1\}$, the N_k circles of shell k occupy *all* the polar angles $\pi(n+\frac{1}{2})/N_k$, and every direction is within $\pi/(2N_k)$ in angle of one of them. In the meridian half-plane $(s,z) = (r\sin\theta, r\cos\theta)$ the circles are points forming a sheared rectangular lattice: L_k layers at radial spacing $2^k/L_k \in \{s_k, s_k/\sqrt{2}\}$, each containing M_k circles at angular spacing $\pi L_k/N_k$, that is, at physical spacing between $0.79\,s_k$ and $2.2\,s_k$; consecutive layers are offset by one angular step π/N_k (Figure 1a). Let $\Gamma = \bigcup_{k,i,j} \Gamma_{k,i,j}$ and define, for $\alpha \geq 4$,

$$w(x) = \Big(\frac{1 + d(x)^2}{1 + |x| + d(x)^2}\Big)^{\alpha/2}, \qquad d(x) = \operatorname{dist}(x, \Gamma). \tag{3}$$

Write $s(r)^2 = 1 + r$. Near a circle at radius r, $w \asymp (1 + d^2)^{\alpha/2}/s^\alpha$ for $d \leq s/2$: the cylinder weight of Theorem 3.4 normalized by the local spacing. For $d \geq s$, w is bounded below by $2^{-\alpha/2}$. The set where w is small is therefore a union of thin tubes around circles that are $\asymp s_k$ apart; no surface of small weight exists at any scale. Two features of the construction are forced. The spacing $s_k \asymp 2^{k/2}$ comes from counting: $N_k \asymp 2^k$ circles of length $\asymp 2^k$ must be spread through a shell of

volume $\asymp 8^k$ at mutual distance s, so $s^2 \asymp 8^k/4^k$. The exponent $\alpha \geq 4$ is the threshold at which a ray that misses a circle by a bounded distance still pays a constant per shell, see (4).

The function $\log w$ is Lipschitz: $\nabla \log w = \frac{\alpha}{2}\big[2d\nabla d/(1+d^2) - (\nabla|x| + 2d\nabla d)/(1+|x|+d^2)\big]$ with $|\nabla d| \leq 1$ almost everywhere, so $|\nabla \log w| \leq \frac{\alpha}{2}(1+1.62)$, and in fact $\sup|\nabla \log w| = \alpha/2 + o(1)$. By Lemma 2.3, $f^* = \varphi * w$ is a smooth Gaussian marginal comparable to w with drift bounded by $\alpha/2 + o(1)$.

Theorem 4.1 (the lattice prior)**.** *Let w be the weight (3) on $\mathbb{R}^3$, $\alpha \geq 4$, and $F = w\,\mathrm{d}\theta$. Then:*

(a) *$w\,\mathrm{d}x$ is doubling and supports the scale-invariant Poincaré inequality (1) on all balls;*
(b) *Brown's sphere test is silent, $\bar{f}(r) \geq c > 0$ for $r \geq 16$; Brown's ray test is silent, and in fact $\int_{16}^{\infty} \mathrm{d}r/(r^2 w(r\varphi)) = \infty$ for* every *direction φ;*
(c) *$V(r) \asymp r^3$.*

Consequently $(\mathbb{R}^3, f^*\,\mathrm{d}x)$ is transient and δ_F is inadmissible, although neither of Brown's radial tests applies.

Proof of (b), (c) and of the consequence. Fix a direction φ and a shell k. Some θ_n is within $\pi/(2N_k)$ of the polar angle of φ, so the ray passes within distance $r\pi/(2N_k) \leq \pi/4$ of a circle of shell k at some radius $r_0 \leq 2^{k+1}$. For $r \in [r_0 - 1, r_0 + 1]$ one has $d^2 \leq 1 + \pi^2/16 < 1.62$ and hence $w \leq (2.62/r_0)^{\alpha/2}$, so

$$\int_{r_0-1}^{r_0+1} \frac{\mathrm{d}r}{r^2 w(r\varphi)} \;\geq\; \frac{2\,(r_0/2.62)^{\alpha/2}}{(r_0+1)^2} \;\geq\; 2.62^{-\alpha/2}\, r_0^{\alpha/2-2}, \tag{4}$$

using $2(r_0/(r_0+1))^2 \geq 1$ for $r_0 \geq 16$. The coefficient $2.62^{-\alpha/2}$ depends on α: the per-shell toll is a positive constant (at least 0.14) for $\alpha = 4$ and grows like $2^{k(\alpha/2-2)}$ for $\alpha > 4$. Summing over k the ray integral diverges, for every φ. For the sphere test, on each sphere $\{|x| = r\}$ with r in shell k the set of points at distance $\geq s_k/4$ from Γ has a surface-measure fraction bounded below by a constant c_0 (the tubes of radius $s_k/4$ around circles that are at least $0.7\,s_k$ apart cover a bounded fraction of any sphere); there $w \geq \big((s_k/4)^2/(1 + r + s_k^2/16)\big)^{\alpha/2} \geq 17^{-\alpha/2}$. So $\bar{f}(r) \geq c_0 17^{-\alpha/2}$ and the sphere integral $\int \mathrm{d}r/(r^2\bar{f})$ converges. For (c), $w \leq 1$ gives $V(r) \leq \frac{4}{3}\pi r^3$, and the same fraction argument applied to shells gives $V(r) \geq cr^3$. Finally, by (a) and (c), $\int^{\infty} r\,\mathrm{d}r/V(r) < \infty$, so $(\mathbb{R}^3, w\,\mathrm{d}x)$ is transient by Theorem 2.4; by Lemma 2.3(iv) so is $(\mathbb{R}^3, f^*\,\mathrm{d}x)$, and δ_F is inadmissible by Theorem 2.1(b). □

4.2. **Proof of (a): doubling and Poincaré.** The proof assembles the Poincaré inequality on a large ball from the inequality on cells containing at most one circle, where the weight is the cylinder weight of Theorem 3.4 up to constants. We first record the assembly lemma and then the inequality for one circle.

Lemma 4.2 (gluing)**.** *Let μ and ν be measures on $\mathbb{R}^n$ with ν D+P, and let $B = B(x_0, \rho)$. Suppose that $2B$ is covered by cubes Q_j with bounded overlap and side lengths $s_j \leq \rho$ such that $s_j \asymp s_{j'}$ whenever $Q_j \cap Q_{j'} \neq \emptyset$, and that for each j there is a ball $B_j \supset \bigcup_{Q_{j'} \cap Q_j \neq \emptyset} Q_{j'}$ of radius $\leq C s_j$, the balls B_j having bounded overlap in $4B$, such that*

(G1) *$\mu(Q_j) \asymp \nu(Q_j)$ for all j, and*
(G2) *μ is doubling at scales $\leq C s_j$ around B_j and satisfies (1) on B_j,*

all with constants independent of j and of B. Then $\int_B |u - u_B|^2\,\mathrm{d}\mu \leq C'\rho^2 \int_{4B} |\nabla u|^2\,\mathrm{d}\mu$ with C' depending only on the constants above.

Proof. Let $\bar{u}_j$ be the μ-mean of u on Q_j, let $\{\psi_j\}$ be a partition of unity subordinate to slight enlargements of the Q_j with $|\nabla\psi_j| \lesssim 1/s_j$, and let $U = \sum_j \bar{u}_j \psi_j$. For any constant m, $\int_B |u - m|^2\,\mathrm{d}\mu \leq 2\int_B |u - U|^2\,\mathrm{d}\mu + 2\int_B |U - m|^2\,\mathrm{d}\mu$. On Q_j, $|u - U| \leq \sum_{j' \sim j} \psi_{j'} |u - \bar{u}_{j'}|$, and by (G2)

on $B_{j'} \supset Q_j$, $\int_{Q_j}|u-\bar u_{j'}|^2\,\mathrm{d}\mu \lesssim s_j^2\int_{B_{j'}}|\nabla u|^2\,\mathrm{d}\mu$; summing with bounded overlap, the first term is $\lesssim \rho^2\int_{4B}|\nabla u|^2\,\mathrm{d}\mu$. For the second term, on Q_j one has $|U-m|^2 \le 2|\bar u_j - m|^2 + 2|U-\bar u_j|^2$ and $|U-\bar u_j| \le \sum_{j'\sim j}|\bar u_{j'}-\bar u_j|$, so by (G1)

$$\int_B |U-m|^2\,\mathrm{d}\mu \;\lesssim\; \sum_j \nu(Q_j)|\bar u_j - m|^2 + \sum_j \nu(Q_j)\sum_{j'\sim j}|\bar u_{j'}-\bar u_j|^2.$$

Choosing m to be the ν-mean of U on $2B$ and using $\nu(Q_j)|\bar u_j - m|^2 \le 2\int_{Q_j}|U-m|^2\,\mathrm{d}\nu + 2\nu(Q_j)|U-\bar u_j|^2$ together with the Poincaré inequality for ν on $2B$ and $|\nabla U|^2 \lesssim s_j^{-2}\sum_{j'\sim j}|\bar u_{j'}-\bar u_j|^2$ on Q_j, everything is bounded by $\rho^2\sum_j \nu(Q_j)s_j^{-2}\sum_{j'\sim j}|\bar u_{j'}-\bar u_j|^2$ (the factor ρ^2 absorbs the second sum since $s_j \le \rho$). Finally, for $Q_{j'}\cap Q_j \neq \emptyset$ both cubes lie in B_j, and by (G2) $|\bar u_{j'}-\bar u_j|^2 \lesssim \mu(Q_j)^{-1}\int_{B_j}|u-u_{B_j}|^2\,\mathrm{d}\mu \lesssim s_j^2\mu(Q_j)^{-1}\int_{B_j}|\nabla u|^2\,\mathrm{d}\mu$ (local doubling makes $\mu(B_j)\lesssim\mu(Q_j)$). With $\nu(Q_j)\asymp\mu(Q_j)$ the factors $s_j^{\pm 2}$ and $\mu(Q_j)^{\pm 1}$ cancel, and bounded overlap of the B_j gives the claim. □

Under doubling of μ the inequality with energy over $4B$ self-improves to (1) [12, Thms. 5.1 and 9.7]; doubling of μ at scales above the cell size follows from (G1) and doubling of ν, and at smaller scales from (G2). We also use that D+P is invariant under bi-Lipschitz changes of variables whose Jacobian is bounded above and below: images of balls are sandwiched between balls of comparable radii, which gives the weak form of (1), and the same self-improvement applies.

Lemma 4.3 (one circle)**.** *Let $C_R \subset \mathbb{R}^3$ be a circle of radius $R \ge 0$ and $w_R(x) = (1+\operatorname{dist}(x,C_R)^2)^{\alpha/2}$, $\alpha \ge 0$. Then $w_R\,\mathrm{d}x$ is D+P with constants independent of R.*

Proof. Scales $\rho \le R/2$. Let $B = B(x_0,\rho)$ with $\operatorname{dist}(x_0,C_R) \le 2\rho$ (otherwise w_R varies on B by a bounded factor and μ is comparable to Lebesgue measure there). In Fermi coordinates (t,ν) along C_R, arclength t and normal vector ν with $|\nu| < R$, the map that sends (t,ν) to the point with the same coordinates relative to a straight line is bi-Lipschitz with constant ≤ 2 on $\{|\nu| \le R/2\}$, has Jacobian $1-\nu_{\mathrm{in}}/R \in [\frac12,\frac32]$, and preserves $\operatorname{dist}(\cdot,C_R) = |\nu|$. It carries w_R on a neighbourhood of B to the cylinder weight $(1+|x'|^2)^{\alpha/2}$ of Theorem 3.4, which is D+P by [3, Prop. 6]; the invariance just noted gives (1) on B.

Scales $\rho \ge R/4$. Let c be the centre of C_R and $p_c(x) = (1+|x-c|^2)^{\alpha/2}$, which is D+P by [3, Prop. 6] with $k = n = 3$. We apply Lemma 4.2 with $\nu = p_c\,\mathrm{d}x$ and cubes of side $R/4$ (or side 1 if $R < 4$, in which case the weight is bounded above and below on cells and nothing is to prove). (G2) holds at scales $\le R/2$ by the first part. For (G1) we show $\mu_R(B') \asymp \nu(B')$ for every ball $B' = B(x,\rho)$ with $\rho \ge R/4$, which also gives doubling of μ_R at those scales. Write $D = |x-c|$. Both measures are $\lesssim \rho^3\max(1,\rho,D)^\alpha$ since $\operatorname{dist}(y,C_R) \le |y-c| + R \le D+\rho+4\rho$ on B'. For the lower bound, if $D \ge 8\rho$ then $\operatorname{dist}(y,C_R) \ge D-\rho-R \ge D/2$ on B'; if $D \le 8\rho$ then $R \le 4\rho$ and the tube $\{\operatorname{dist}(\cdot,C_R) \le \varepsilon\rho\}$ has volume $\lesssim (R+\varepsilon\rho)(\varepsilon\rho)^2 \le C\varepsilon^2\rho^3$, so for ε small a fixed fraction of B' lies at distance $\ge \varepsilon\rho$ from C_R, where $w_R \ge (\varepsilon\rho)^\alpha \gtrsim \max(1,\rho,D)^\alpha$ (for $\rho \ge 1$; for $\rho < 1$ all weights involved are $\asymp 1$). This is the lower bound in both cases. □

Proof of Theorem 4.1(a). Let $B = B(x_0,\rho)$. If $\rho \le s(x_0)/4$ and $2^k \le |x_0| < 2^{k+1}$, the ball meets at most one circle, because distinct circles are at distance $\ge 0.7\,s_k$ and $s(x) \le \sqrt{2}\,s_k$ on B. If no circle lies within distance ρ of B, then $d \ge \rho$ on B and w varies on B by a factor at most $\sup_u W(3u)/W(u) \le 9^{\alpha/2}$, where $W(u) = (u^2/(1+u^2))^{\alpha/2}$ is the profile of w in the variable $u = d/s$; so μ is comparable to Lebesgue measure on B and (1) holds. Otherwise B lies in a ball B' of radius $\le s_k$ around a point of one circle C, and on B' the weight equals $(1+d^2)^{\alpha/2}$ times $(1+|x|+d^2)^{-\alpha/2}$, where the second factor varies on B' by a factor $1+O(1/s_k)$ (both $|x|$ and d^2 change by $O(s_k)$ against $1+|x| \ge 4^k$ and $d \le s_k$); Lemma 4.3 applies to the first factor. Doubling at these scales follows in the same way.

If $\rho > s(x_0)/4$, apply Lemma 4.2 with ν the Lebesgue measure and a covering of $2B$ by cubes Q_j of side $s(y_j)/8$, y_j the centre of Q_j; adjacent cubes have comparable sides since s varies by a bounded factor over distances $\le s$. (G2) at scales $\le s(y_j)/4$ is the case just treated. For (G1) we need $\mu(Q_j) \asymp |Q_j|$ uniformly. In the variable $u = d/s$ the weight is $W(u)(1 + O(1/s))$, and the geometry of the circles relative to a cube of side $s/8$ is one of finitely many patterns up to the $O(1/s)$ distortion caused by the curvature of the circles and by the change of spacing across shell boundaries: a cube either meets no circle, in which case $u \ge$ some $u_0 > 0$ on it and $W \ge W(u_0)$, or meets one, in which case the mean of $W(d/s)$ over the cube is bounded below by a constant depending on α only (the tube $\{u \le \varepsilon\}$ occupies a fraction $\lesssim \varepsilon^2$ of the cube). Since $W \le 1$, $\mu(Q_j) \asymp |Q_j|$. Now Lemma 4.2 gives the weak Poincaré inequality on B, doubling at scale $\ge s/8$ follows from (G1), and the self-improvement gives (1). □

Remark 4.4. The prior $F = w\,\mathrm{d}\theta$ is merely Lipschitz-logarithmic (the distance to a union of circles is not differentiable on its medial surfaces); its marginal $f^* = \varphi * w$ is C^∞, and by Lemma 2.3 it inherits every property proved for w. The construction is axisymmetric; this is a convenience for the numerics of Section 6 and plays no role in the proofs.

5. The zone is two-sided

Theorem 4.1 exhibits the one kind of case that the volume test decides inside the D+P class. Outside the class the undecided zone contains members of both types, decided by other non-radial arguments. The first example also shows that the Poincaré hypothesis in Theorem 2.4 cannot be dropped, and explains why the circles of Section 4 had to be scattered.

5.1. **Coherent circles form a wall.** Place, for $k \ge 4$ and $0 \le j < 2^k$, the circle $\Gamma_{k,j} = \{|x| = 2^k + j,\ \theta = \pi(j + \frac12)/2^k\}$, let $\Gamma' = \bigcup \Gamma_{k,j}$, and let

$$w'(x) = \Big(\frac{1 + d'(x)^2}{1 + |x|^2}\Big)^{\alpha/2}, \qquad d' = \operatorname{dist}(\cdot, \Gamma'), \quad \alpha \ge 2. \tag{5}$$

Every ray meets a tube of bounded radius around Γ' in every shell, so the ray test is silent, and $\bar f$ is bounded below, so the sphere test is silent; $\log w'$ is Lipschitz. One might expect transience from the volume growth $V(r) \asymp r^3$. The opposite is true.

Lemma 5.1 (flux bound along a non-spherical exhaustion)**.** *Let w be a positive weight on $\mathbb{R}^m$ and let $\{S_t\}_{t\in T}$ be pairwise disjoint regions, each fibred by nested closed Lipschitz surfaces $S_t^{(s)}$, $s \in [s_0, s_1]$, all enclosing B_1, with volume element $\mathrm{d}V = \cos(\text{tilt})\,\mathrm{d}A\,\mathrm{d}s$ along the fibration. Then every vector field J with $\operatorname{div} J = 0$ outside B_1 and unit flux out of B_1 satisfies*

$$\int \frac{|J|^2}{w}\,\mathrm{d}V \;\ge\; \sum_{t\in T} \int_{s_0}^{s_1} \frac{\mathrm{d}s}{\tilde F_t(s)}, \qquad \tilde F_t(s) = \int_{S_t^{(s)}} \frac{w}{\cos(\text{tilt})}\,\mathrm{d}A.$$

If the right-hand side is infinite, then $\operatorname{cap}(B_1) = 0$ and $(\mathbb{R}^m, w\,\mathrm{d}x)$ is parabolic.

Proof. The flux of J through each $S_t^{(s)}$ equals 1. By the Cauchy–Schwarz inequality, $1 = \big(\int J \cdot n\,\mathrm{d}A\big)^2 \le \int |J|^2 \cos(\text{tilt}) w^{-1}\,\mathrm{d}A \cdot \tilde F_t(s)$; integrate in s with $\mathrm{d}V = \cos(\text{tilt})\,\mathrm{d}A\,\mathrm{d}s$ and sum over the disjoint regions. By Thomson's principle, the dual of the Dirichlet principle in Theorem 2.1(a), $\operatorname{cap}(B_1)$ is the reciprocal of the minimal energy of such flows [8, §7]. □

Proposition 5.2 (the ring weight is recurrent)**.** *For the weight (5) with $\alpha \ge 2$, $(\mathbb{R}^3, w'\,\mathrm{d}x)$ is parabolic, and δ_F is admissible for $F = w'\,\mathrm{d}\theta$. The weight is doubling but does not support the scale-invariant Poincaré inequality.*

Proof. The circles of shell k lie on the surface of revolution $\Sigma_k:\ r=\rho_k(\theta)=2^k(1+\theta/\pi)-\frac12$, $\theta\in[0,\pi]$, a closed graph over the sphere running from $r=2^k-\frac12$ at the north pole to $r=2^{k+1}-\frac12$ at the south pole; consecutive surfaces reach the common radius $2^{k+1}-\frac12$ at antipodal poles, not at a shared point. Along Σ_k consecutive circles are at distance between 3.3 and 6.4 (radial step 1, meridian step $r\pi/2^k\in[\pi,2\pi]$), so every point of Σ_k is within 3.2 of Γ'. On the layer $\{r=\rho_k(\theta)+s,\ |s|\le1\}$ therefore $d'\le4.2$ and $w'\le(19/4^k)^{\alpha/2}$, while the weighted area $\tilde F_k(s)\le(19/4^k)^{\alpha/2}\,C\,4^k$ since $\mathrm{Area}(\Sigma_k)\asymp4^k$ and $\cos(\mathrm{tilt})\ge0.95$. Apply Lemma 5.1 to the layers of the even shells, which are pairwise disjoint: each contributes $\int_{-1}^{1}\mathrm{d}s/\tilde F_k(s)\ge c\,4^{k(\alpha/2-1)}$, a constant for $\alpha=2$ and geometrically growing for $\alpha>2$. The sum diverges, so $\mathrm{cap}(B_1)=0$. Admissibility follows from Theorem 2.1(b) and Lemma 2.3. For the Poincaré failure take a ball B of radius ρ centred on Σ_k at $\theta=\pi/2$ with $4\ll\rho\ll2^k$, and the two-region test function $u=\min(1,\max(0,(s+1)/2))$ in the signed normal distance s to Σ_k. On either side of the wall $w'\asymp(|s|/2^k)^\alpha$, so $\mathrm{Var}_\mu(u;B)\asymp\rho^3(\rho/2^k)^\alpha$, whereas the energy is carried by the layer $|s|\le1$ and is $\asymp\rho^2\,4^{-k\alpha/2}$. Hence $\mathrm{Var}_\mu(u;B)/(\rho^2\int_B|\nabla u|^2\,\mathrm{d}\mu)\asymp\rho^{\alpha-1}$ is unbounded. Doubling holds because $V(x,\rho)\asymp\rho^3\cdot(\max(\rho,\mathrm{dist}(x,\Sigma))/2^k)^\alpha$ at the relevant scales. □

Above its mesh scale the set Γ' is a surface of small weight: a valley of co-dimension one, the obstruction of Lemma 3.3 in three dimensions. Theorem 3.1 says in advance that such a weight could not have been D+P: it is parabolic with the sphere test silent.

5.2. **A logarithmic product prior.**

Proposition 5.3 (product prior)**.** *Let $F(\,\mathrm{d}\theta)=v(\theta_1)\,\mathrm{d}\theta_1\,\mathrm{d}\theta_2\,\mathrm{d}\theta_3$ on $\mathbb{R}^3$ with $v(t)=(1+|t|)^{-1}\log(e+|t|)$. Then both of Brown's tests are silent, v is not doubling, and $(\mathbb{R}^3,f^*\,\mathrm{d}x)$ is transient, so δ_F is inadmissible.*

Proof. $\log v$ is Lipschitz, so we may work with $w=v(x_1)$. The sphere mean is $\bar f(r)\asymp\mu_1([-r,r])/r\asymp\log^2 r/r$, and $\int\mathrm{d}r/(r^2\bar f)$ converges. Along a direction with $\cos\varphi\neq0$ the ray integral equals $|\cos\varphi|$ times $\int^\infty\mathrm{d}r/(r\log r)=\infty$, and the equatorial directions form a null set, so the ray test is silent. The doubling ratio $\mu_1([0,2r])/\mu_1([r/2,3r/2])\asymp\log r$ is unbounded. For transience, the weighted Brownian motion is the product of the one-dimensional diffusion W with generator $v^{-1}(vu')'$ and a standard Brownian motion on $\mathbb{R}^2$, and it is transient if and only if $\int^\infty p_t^W(0,0)\,t^{-1}\,\mathrm{d}t<\infty$, where p^W is the transition density with respect to $v\,\mathrm{d}t$. In natural scale, $s(x)=\int_0^x\mathrm{d}t/v\sim x^2/(2\log x)$, the speed measure is $\widetilde M(s)=\int_0^{x(s)}v\sim\frac18\log^2 s$, which is slowly varying. The resolvent at the origin, $h(\lambda)=\int_0^\infty e^{-\lambda t}p_t^W(0,0)\,\mathrm{d}t$, is the characteristic function of the string $\widetilde M$ in Krein's correspondence [15, eqs. (5), (10)], and Kasahara's Tauberian theorem [15, Thm. 2(iii), case $\alpha=1$] (see also [19, 16]) states that $\widetilde M$ is slowly varying at infinity if and only if $h(\lambda)\sim c/(\lambda\widetilde M(1/\lambda))$ as $\lambda\to0$, with an absolute constant c. Karamata's theorem then gives $p_t^W(0,0)\sim c'/\widetilde M(t)\asymp1/\log^2 t$, hence

$$\int^\infty p_t^W(0,0)\,\frac{\mathrm{d}t}{t}\ \asymp\ \int^\infty\frac{\mathrm{d}t}{t\log^2 t}\ <\ \infty.$$

The asymptotic, with constant $c'/\frac18=4$ for the string reflected at the origin, is confirmed numerically in Section 6. □

This is an instance of the phenomenon of Johnstone and Lalley [14, Thm. 1.1]: for every null-recurrent one-dimensional generalized Bayes component there is a product partner for which the product estimator is inadmissible; their construction uses a different one-dimensional factor. Table 2 summarizes the three members of the undecided zone exhibited in this note.

TABLE 2. Three priors on $\mathbb{R}^3$ in Brown's undecided zone. Each has Lipschitz logarithm and silences both radial tests; each is decided by a different non-radial argument.

weight	doubling	Poincaré	W_F; δ_F	decided by
$(1+\lvert x_1\rvert)^{-1}\log(e+\lvert x_1\rvert)$ (Prop. 5.3)	no	yes	transient; inadmissible	product heat kernels
circles on one surface per shell (Prop. 5.2)	yes	no	parabolic; admissible	flux bound along the surfaces (Lemma 5.1)
scattered circles, spacing $2^{k/2}$ (Thm. 4.1)	yes	yes	transient; inadmissible	volume test under D+P (Thm. 2.4)

TABLE 3. Neumann eigenvalues $\lambda_1\rho^2$ (axisymmetric modes; Lebesgue value 4.33). Lattice weight (3) with $\alpha = 4$ on balls centred on the axis at $|x| = 1.5\cdot 2^k$; one-circle weight w_R of Lemma 4.3 on balls centred at the circle's centre; ring weight (5) with $\alpha = 2$ on balls centred on the axis at $z_0 = 256$, which Σ_8 crosses.

weight, ball family	ρ/s or $\rho/R = 1$	2	4	8
lattice, $k = 8$ ($s_8 = 16$)	1.55	2.69	2.93	2.82
lattice, $k = 10$ ($s_{10} = 32$)	1.60	2.71	3.09	3.01
one circle, $R = 1$	2.58	1.45	1.67	2.17
one circle, $R = 4$	1.09	0.53	1.49	2.15
one circle, $R = 16$	0.94	0.47	1.50	2.15
ring weight, $\rho = 8, 16, 32, 64$	0.975	0.497	0.256	0.129

6. Numerical illustration

The proofs above are complete modulo the cited results; the computations in this section were used to find the constructions and to check the uniformity claims that a proof sketch can hide. All numbers are produced by short Python scripts available with this note.

Poincaré constants as eigenvalues. The best constant in (1) on a region is $1/(\lambda_1\rho^2)$, with λ_1 the first nonzero Neumann eigenvalue of the weighted Laplacian. We discretize by finite volumes on a uniform grid (face conductance equal to the geometric mean of the weight on the two adjacent cells, cell mass equal to the weight times the cell volume) and solve the generalized eigenproblem by shift-invert Lanczos. The Lebesgue controls, $\lambda_1 R^2 = \pi^2/4 = 2.467$ on the square $[-R, R]^2$ and $\lambda_1\rho^2 = 4.333$ for the axisymmetric modes on a ball in $\mathbb{R}^3$, are reproduced to three digits. For a ball centred on the symmetry axis of an axisymmetric weight, the azimuthal modes $e^{im\phi}g(s, z)$, $m \geq 1$, satisfy $\int|\nabla u|^2\,\mathrm{d}\mu \geq \int|u|^2 s^{-2}\,\mathrm{d}\mu \geq \rho^{-2}\int|u|^2\,\mathrm{d}\mu$ with mean zero, so they obey (1) with constant 1 for any weight; the $m = 0$ problem on the meridian half-disc, with measure $2\pi s\, w\,\mathrm{d}s\,\mathrm{d}z$, is then the whole statement, and it is a two-dimensional computation that reaches balls of radius 256 at mesh 0.5.

Table 3 and Figure 1b show the outcome. For the lattice weight $\lambda_1\rho^2$ stays between 1.5 and 3.1 from one spacing to eight spacings and across two shells, about two thirds of the Lebesgue value, with no downward trend; the values are converged to three digits under mesh refinement. For the one-circle weight of Lemma 4.3 the eigenvalue is bounded below uniformly in R and in ρ/R, with a dip at $\rho = 2R$ that is the same for $R = 4$ and $R = 16$, as the scaling argument in the proof predicts. For the ring weight of Proposition 5.2 the eigenvalue halves with every doubling of ρ, i.e. decays like $\rho^{-(\alpha-1)}$, the two-region quotient of the proof. For the two-dimensional factor $(1+|y|^2)^2$ of Theorem 3.4, $\lambda_1 R^2$ on $[-R, R]^2$ decreases from 1.01 at $R = 2$ to 0.770 at $R = 64$ and is flat thereafter. Off the axis the reduction is unavailable and a full three-dimensional computation

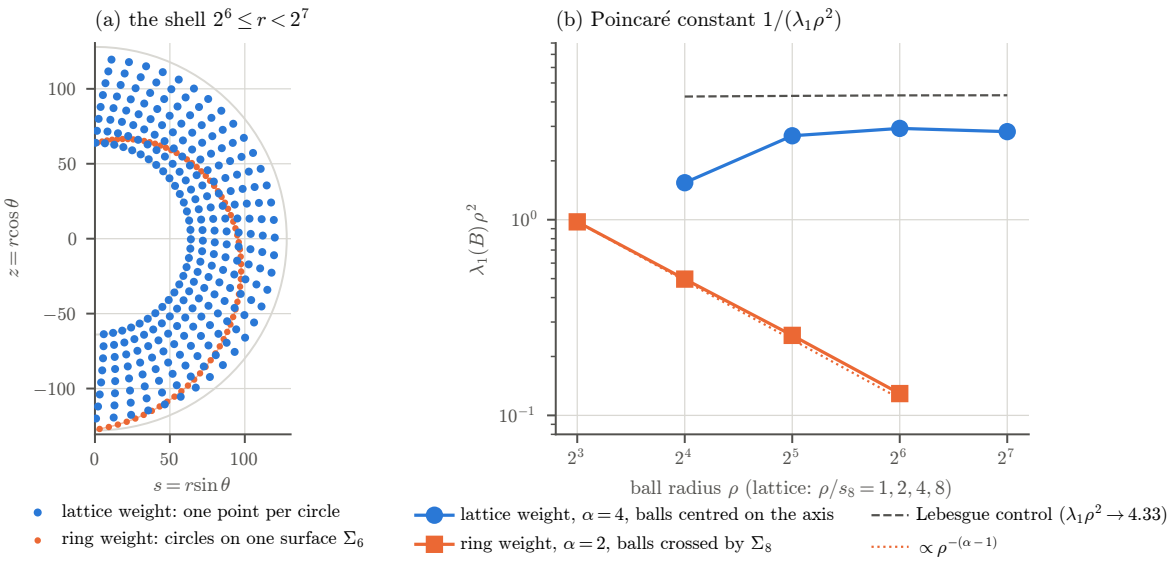


FIGURE 1. (a) Meridian half-plane of the shell $2^6 \le r < 2^7$: the circles of the lattice weight (3) (one point per circle, spacing $\approx s_6 = 8$) and the circles of the ring weight (5), which lie on one surface Σ_6 at mesh ≈ 3. (b) $\lambda_1\rho^2$ on balls centred on the axis: flat for the lattice weight, decaying like $\rho^{-(\alpha-1)}$ for the ring weight.

is needed: on cubes of half-side s_8 and $2s_8$ centred on a point of a mid-shell circle, which contain the strand and its nearest neighbours, $\lambda_1\rho^2 = 0.94$ and 1.16; centred in the bulk between circles, 1.57 and 1.42; centred on the axis, 1.34 and 1.69 (control for a cube, $\pi^2/4 = 2.47$); all values are unchanged to two digits under mesh refinement to $2.6 \cdot 10^5$ cells.

Brown's tests and the volume. For the lattice weight with $\alpha = 4$, the smallest of 802 ray integrals per shell, over random directions, the directions exactly between consecutive θ_n, the poles and the equator, is between 0.94 and 1.92 for $k = 4, \dots, 11$ (the bound (4) gives ≥ 0.14); for $\alpha = 6$ it doubles per shell. The sphere mean lies between 0.0039 and 0.052, and $V(r)/(\frac{4}{3}\pi r^3)$ between 0.019 and 0.029, both alternating with the parity of k. Doubling ratios $\mu(B(x, 2\rho))/\mu(B(x, \rho))$ at centres on a circle, in the bulk and on the axis lie between 4 and 13 for $\rho \ge s_k/4$ and reach the co-dimension-two power-weight values (up to 94) at the tube scale. For the ring weight with $\alpha = 2$ the per-shell flux terms of Lemma 5.1 are flat at 0.047 and for $\alpha = 3$ they double per shell, as in the proof of Proposition 5.2; the two-region quotient divided by $\rho^{\alpha-1}$ is flat at 0.041 ($\alpha = 2$) and 0.013 ($\alpha = 3$) for ρ between 16 and 128.

The one-dimensional heat kernel. For Proposition 5.3 a Crank–Nicolson computation of $p_t^W(0,0)$ (control $v \equiv 1$ reproduces $(4\pi t)^{-1/2}$ to five digits) gives $\log^2 t \cdot p_t^W(0,0) = 2.58, 2.95, 3.18, 3.33, 3.44$ at $t = 10^3, \dots, 10^7$, consistent with the limit 4 of the Tauberian argument (Figure 2a shows the three tests on this weight).

7. Discussion

Brown's 1971 theory is the potential theory of the weighted space $(\mathbb{R}^m, f^*\,\mathrm{d}x)$, and his two radial tests are the sphere-area test and a ray-wise capacity bound. What the later geometric theory adds inside his undecided zone is now delimited. If the marginal is doubling and Poincaré, nothing is added on the admissible side: parabolicity, Brown's sphere test and the volume test coincide (Theorem 3.1). On the inadmissible side the volume test decides weights that no cone of rays decides,

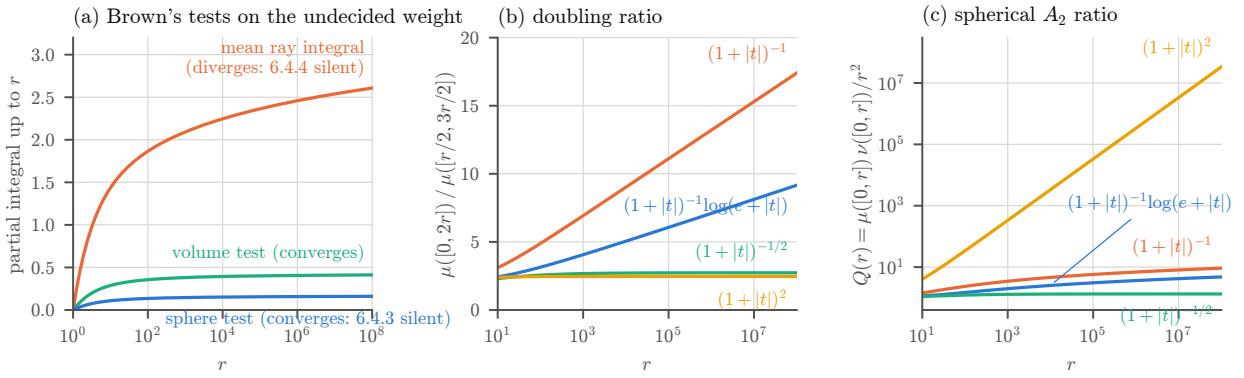


FIGURE 2. (a) Brown's two tests and the volume test on the product weight of Proposition 5.3: the sphere integral and the volume integral converge, the direction-averaged ray integral diverges like $\log\log r$. (b) The doubling ratio $\mu_1([0,2r])/\mu_1([r/2,3r/2])$ of four one-dimensional weights $v(t)$. (c) Their spherical A_2 ratio on $\mathbb{R}^3$: bounded only for $(1+|t|)^{-1/2}$, the one weight that is both doubling and Poincaré (Lemma 3.3); $(1+|t|)^2$ is doubling but fails the Poincaré inequality, $(1+|t|)^{-1}$ is Poincaré but not doubling.

and Theorem 4.1 shows that such weights exist in $\mathbb{R}^3$: valleys of co-dimension two, sweeping every direction at every scale while staying far apart, are exactly what is needed, and within this ansatz the arithmetic of the construction ($N_k \asymp 2^k$ circles, spacing $2^{k/2}$, exponent $\alpha \geq 4$) is forced by the two requirements. Outside the D+P class the zone contains recurrent and transient members, and the first example of Section 5 shows that the Poincaré hypothesis in the volume criterion is not decorative: coherent circles form a wall, and a wall is recurrent with volume growth r^3.

Several questions remain. First, Grigor'yan's flux test [8, Thm. 7.6] already characterizes parabolicity, without the D+P hypothesis, through the existence of a suitable exhaustion (for the smooth positive weights here, via the conformal metric $w^{2/(m-2)}\delta$ when $m \geq 3$); the open question is whether an *explicit, computable* exhaustion, of the kind that decided Proposition 5.2, can be given across the undecided zone. Second, the construction of Theorem 4.1 is three-dimensional; in the plane the valleys would be points, transience requires volume growth faster than $r^2 \log r$, and the arithmetic of exponents changes. Third, the relation between the volume test and the unified sufficient condition of Brown and Hwang [5] for admissibility should be settled; we expect the lattice prior not to be covered by a radial condition of any kind, but we have not checked their hypotheses. Finally, the same parabolic/non-parabolic dichotomy governs Komaki's shrinkage priors on the Fisher manifold of a model [17] (Appendix A); whether the constructions of this note have counterparts there is open.

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Library for access to the primary sources. The scripts that produce every number and figure in Section 6 are available in the public repository `https://github.com/olegroshka/ideg` (directory `articles/theory`).

Appendix A. The same dichotomy for predictive densities

Brown's criterion is not the only place where parabolicity decides a statistical question. Let $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ be a regular parametric model with Fisher metric g, Jeffreys prior $\pi_J \propto |g|^{1/2}$ and Laplace–Beltrami operator Δ_g [1]. For a prior $f = e^{-\Phi}\pi_J$, let $\mathrm{Risk}_N(f)$ be the Kullback–Leibler risk of the Bayesian predictive density built from N observations. Komaki [17] proved that a prior whose predictive density asymptotically dominates the Jeffreys one exists if and only if (M, g) carries a Green function, that is, if and only if the Fisher manifold is non-parabolic (his Theorems 1 and 2; Theorem 3 derives this from negative sectional curvature for $d \geq 3$ through Aomoto's result [2]). For the normal location model the Fisher manifold is flat $\mathbb{R}^d$, the Green function exists iff $d \geq 3$, and the Green prior is Stein's harmonic prior $\|\mu\|^{-(d-2)}$: the predictive analogue of the dichotomy $m \leq 2$ versus $m \geq 3$ of Table 1.

The quantity that Komaki's expansion produces has a name in geometry. The pair (g, Φ) defines the smooth metric measure space $(M, g, e^{-\Phi}\,\mathrm{d}V)$, whose weighted scalar curvature, in Perelman's normalization [21, Remark 1.3], is

$$R^{\mathrm{m}}_{\Phi} = R + 2\Delta_g\Phi - |\nabla\Phi|^2_g, \tag{6}$$

with R the scalar curvature of g; its integral is Perelman's functional $\mathcal{F}(g, \Phi) = \int R^{\mathrm{m}}_{\Phi} e^{-\Phi}\,\mathrm{d}V = \int (R + |\nabla\Phi|^2) e^{-\Phi}\,\mathrm{d}V$.

Proposition A.1. *For a regular model and $f = e^{-\Phi}\pi_J$, with $u = \sqrt{f/\pi_J} = e^{-\Phi/2}$,*

$$\mathrm{Risk}_N(\pi_J) - \mathrm{Risk}_N(f) = \frac{2\Delta_g\Phi - |\nabla\Phi|^2_g}{2N^2} + o(N^{-2}) = \frac{R^{\mathrm{m}}_{\Phi} - R}{2N^2} + o(N^{-2}) = -\frac{2}{N^2}\,\frac{\Delta_g u}{u} + o(N^{-2}). \tag{7}$$

Hence f asymptotically dominates π_J at this order if and only if u is a positive superharmonic function on (M, g) that is not harmonic (equivalently $R^{\mathrm{m}}_{\Phi} \geq R$, with strict inequality on a set of positive measure). A nonconstant positive harmonic *u makes the $O(N^{-2})$ coefficient vanish, so nonconstancy alone is not enough; the existence of such a superharmonic u is, however, equivalent to non-parabolicity of (M, g).*

Proof. The first equality is Komaki's expansion [17, eqs. (3)–(5)]: in the difference of the risks of two priors the prior-independent terms and the skewness terms cancel. The second is the definition (6), and the third is the identity $2\Delta\Phi - |\nabla\Phi|^2 = -4\Delta u/u$ for $u = e^{-\Phi/2}$. The domination statement is Komaki's Theorem 2. □

So the operator that governs domination is $-4\Delta_g$ acting on u, Perelman's operator $-4\Delta + R$ with the potential removed. The scalar curvature itself has no statistical counterpart:

Proposition A.2. *The prior-independent $O(N^{-2})$ term of the risk of the Jeffreys predictive density is not proportional to R, and $\mathcal{F}$ is not the integrated risk reduction: whenever the boundary term of one integration by parts vanishes,*

$$\begin{aligned}\int \big(\mathrm{Risk}_N(\pi_J) - \mathrm{Risk}_N(f)\big) e^{-\Phi}\,\mathrm{d}V &= \frac{1}{2N^2}\int |\nabla\Phi|^2_g e^{-\Phi}\,\mathrm{d}V + o(N^{-2}) \\ &= \frac{1}{2N^2}\Big[\mathcal{F}(g, \Phi) - \int R\, e^{-\Phi}\,\mathrm{d}V\Big] + o(N^{-2}).\end{aligned} \tag{8}$$

Proof. Komaki [18, Theorem A1, eq. (29)] gives the full $O(N^{-2})$ risk of the Jeffreys predictive density. Its prior-independent part is a fixed combination of the exponential and mixture embedding curvatures, the squared norm of the skewness tensor, a fourth-moment contraction and one mixed

moment. For $d = 1$ these are generically nonzero, Efron's statistical curvature of a curved exponential family being the simplest instance [7], while every intrinsic curvature of a one-dimensional Riemannian manifold vanishes. Equation (8) follows from (7) and $\int \Delta_g \Phi \, e^{-\Phi} \, \mathrm{d}V = \int |\nabla \Phi|^2 e^{-\Phi} \, \mathrm{d}V$. The integrated form is an identity of leading coefficients; integrating the pointwise $o(N^{-2})$ remainder against $e^{-\Phi} \, \mathrm{d}V$ requires in addition a dominated or uniformly integrable remainder bound, which we do not establish for improper Φ. □

Two remarks complete the parallel. First, Komaki's treatment of vague priors [17, §3] decides domination over $h = e^{-\Phi}\pi_J$, $d \geq 3$, by superharmonicity with respect to the conformal metric $\tilde{g} = (h/\pi_J)^{2/(d-2)} g$: changing the reference prior is a conformal change of the Fisher metric, and for Stein's prior on $\mathbb{R}^d$ the conformal metric is the round cylinder $S^{d-1} \times \mathbb{R}$, of scalar curvature $(d-1)(d-2)$, with the origin pushed to infinite distance (his Example 1); it is parabolic despite that positive curvature. Second, in both theories the geometry of the prior enters through recurrence or transience of a diffusion, Brown's on the sample space and Komaki's on the parameter space, and not through curvature in a risk formula. On $\mathbb{R}^3$ the scattered valleys of Theorem 4.1 are decided by the volume criterion under $D+P$ although neither of Brown's one-sided radial tests fires (both the sphere and the volume integrals converge there). Whether such weights have an analogue on a Fisher manifold, where Komaki's criterion would then separate the existence of a shrinkage prior from the sphere-area growth of the model, is open.

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